Taking Flight: Using NASA's G-LiHT Airborne Data to Illuminate Carbon Cycling in Mid-Atlantic Ecosystems

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Fellowship of the Wing

Overlap Periods

Period	B200-FX	P-3	G-III	B200-AV	TO	A90
June 22 – 26	x	x				
July 7 - 15	x	x	x	x	x	
July 16 - 31		x	x	x	x	
August			x			x

B200-FX @ 0.5 – 10 kft
Gases + fluxes
In situ

TO @ 1 – 10 kft
Gases + wind profiler
In situ

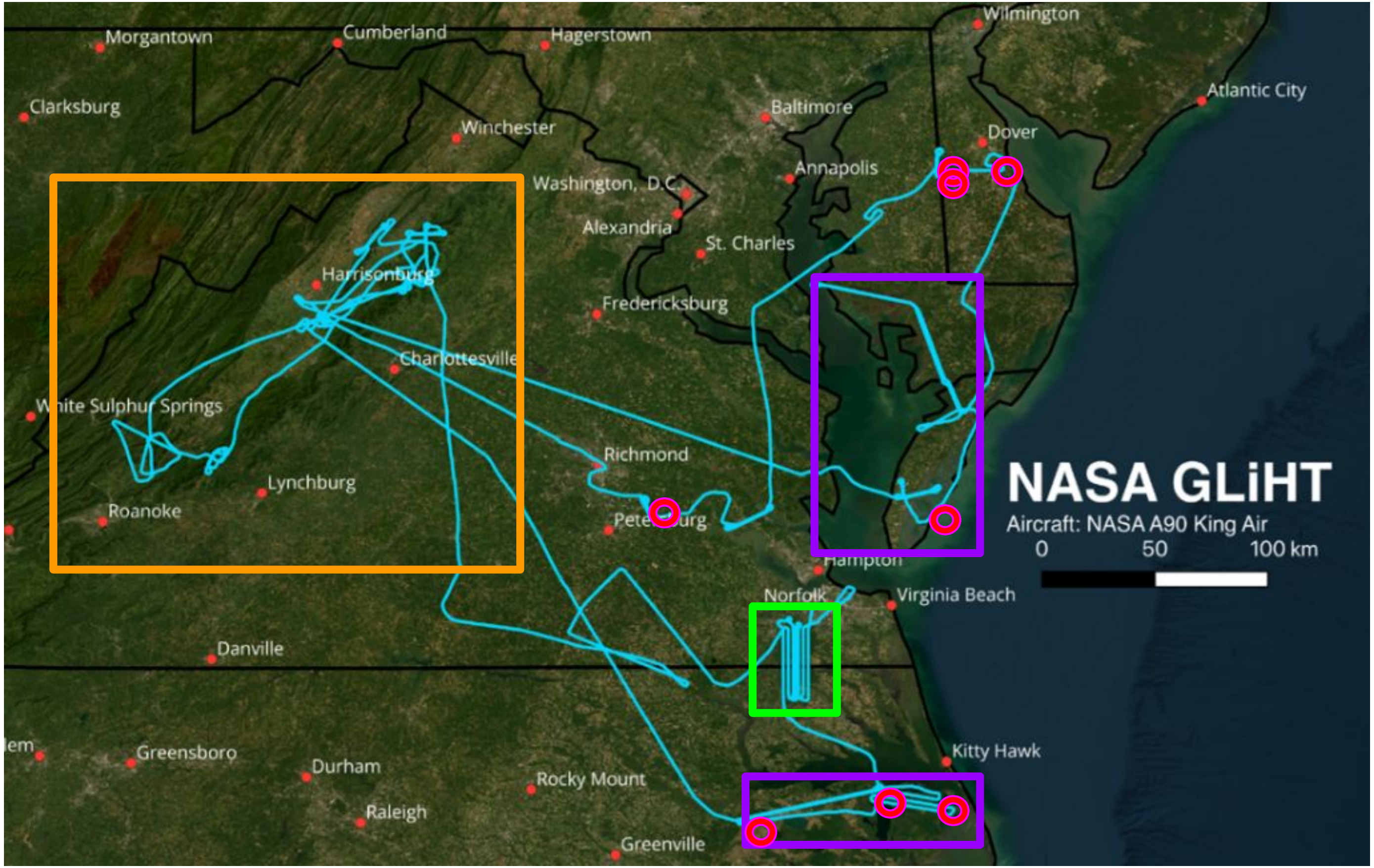
P3 @ 0 – 25 kft
Gas + aerosol
In situ

A90 @ 1 kft
Vegetation
remote sensing

B200-AV @ 28 kft
Atmos + ground
remote sensing

GIII @ 42 kft
Atmosphere
remote sensing

2025 MAGEQ G-LiHT Flight Paths



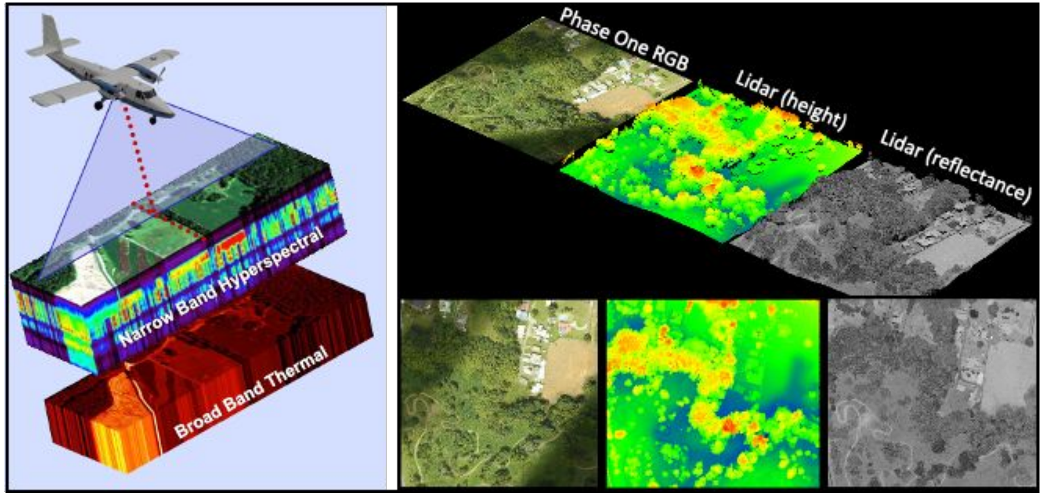
Appalachian Ecosystems (Fire History)

Coastal Ecosystems NC/MD (Sea level rise, Restoration)

Great Dismal Swamp NWR (G-LiHT/CARAFE testbed)

MidFlux Wetland EC Towers

Sensor Platform Description



Goddard's
Lidar
Hyperspectral
Thermal

3D Imaging of Earth's Biosphere

Dual scanning lidar (Riegel VQ-480i)
- 1550 nm laser, density up to 12pt/m^2

Co-aligned Headwall Imaging Spectrometers
- Spectral range: 400 to 2500 nm (VSWIR)

Multiple thermal imaging (TIR) options

PhaseOne IXU-R 1000
- NIR & RGB
- 100 Megapixel, high frame rate stereo-imaging

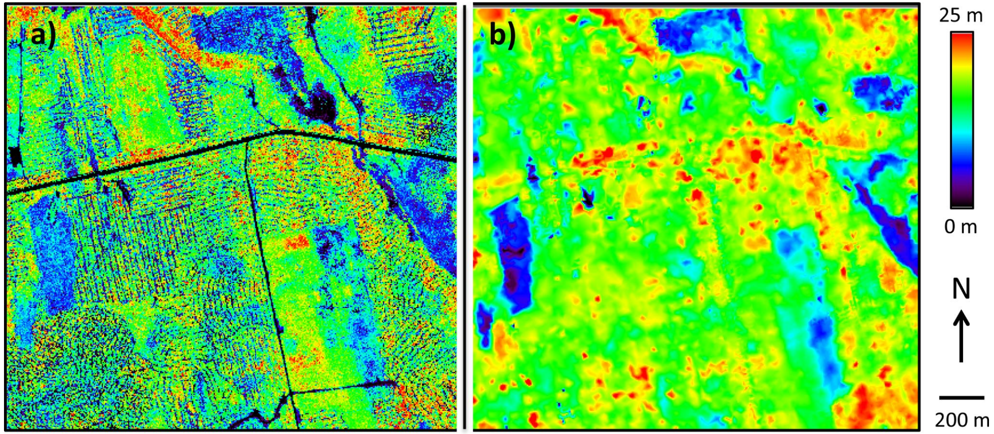
King Air A-90 (Dynamic Aviation)

Data use cases in Ecosystem Carbon Science

- Small footprint lidar/ stereo-imaging: fine scale canopy structure related to management/disturbance (right).
- VSWIR hyperspectral: species composition, function and ecophysiology (2^o pigments)
- Thermal: ET and stress

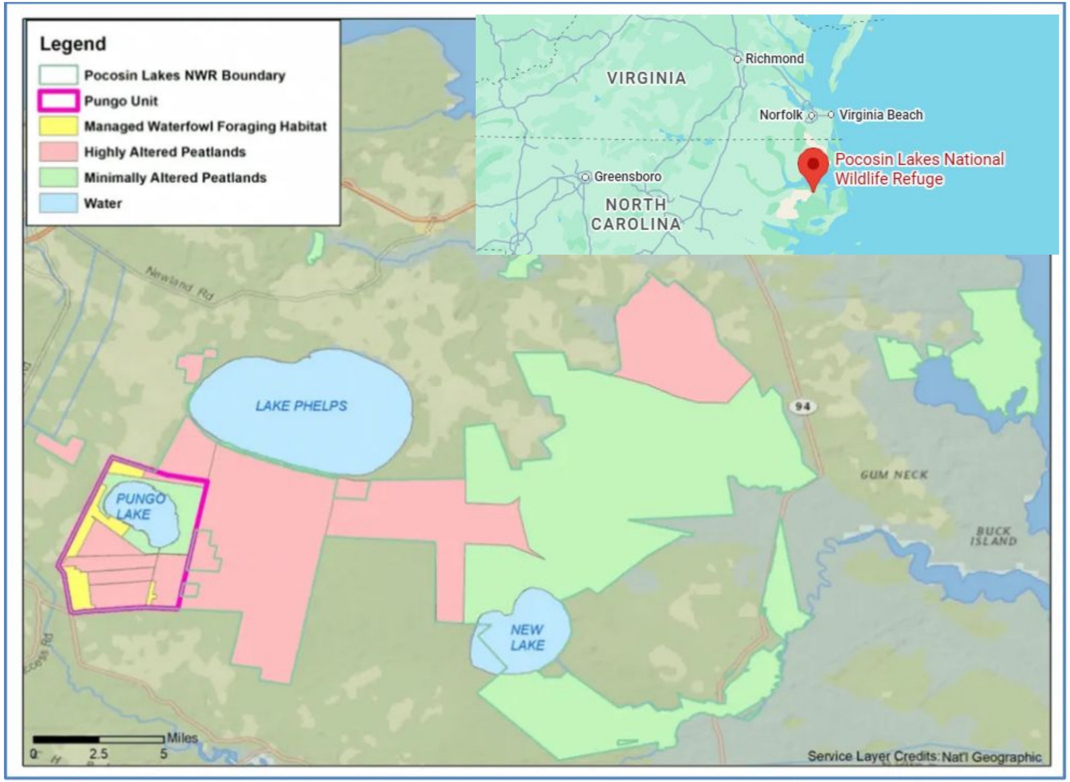
Applications:

- growth/recovery (repeat)
- stress gradients (transect)
- phenology/extreme events
- structure-flux relationships (integration)

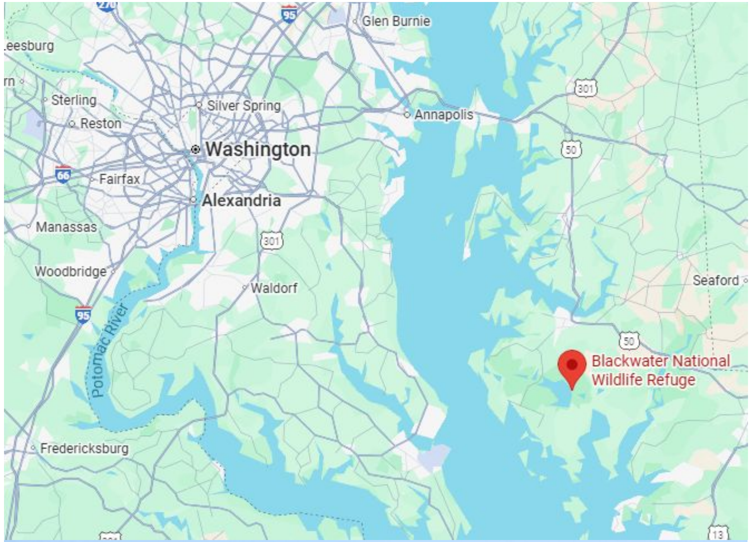


Canopy height model from (a) small footprint (10 cm) G-LiHT LiDAR during June 2012; and (b) large footprint (25 m) Land, Vegetation and Ice Sensor (LVIS) LiDAR during August 2009 [17], for a commercial forest near Howland, ME, USA (45.2220°N 68.7423°W). Discrete returns from small footprint LiDAR are able to detect small gaps and characterize fine-scale disturbance (i.e., strip harvesting), which are challenging to deconvolve from large footprint LiDAR waveforms. From: Cook, B.D., et al., 2013. NASA Goddard's LiDAR, hyperspectral and thermal (G-LiHT) airborne imager. Remote Sensing, 5(8), pp.4045-4066.

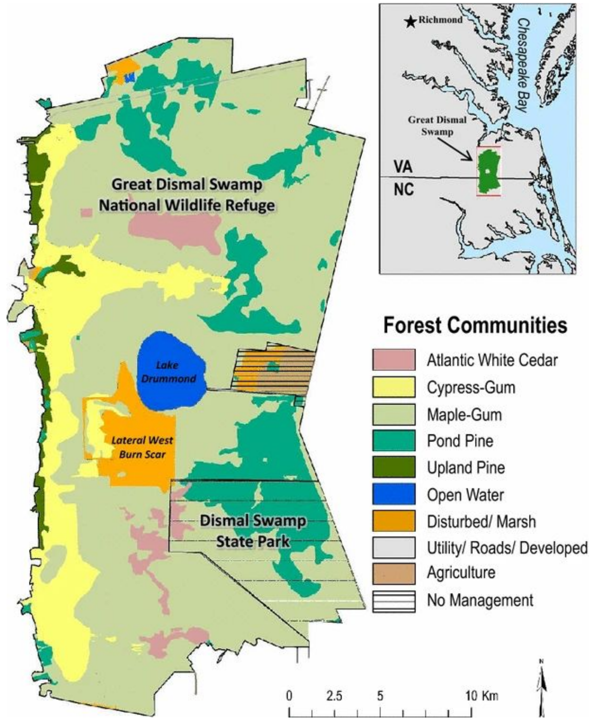
Wetland Ecosystems NC/MD (Sea Level Rise, Restoration)



Map 1: Pocosin Lakes National Wildlife Refuge Management Areas (U.S.FWS 2020)



Great Dismal Swamp (G-LiHT/CARAFE testbed)



Sleeter, R., Sleeter, B.M., Williams, B., Hogan, D., Hawbaker, T. and Zhu, Z., 2017. A carbon balance model for the great dismal swamp ecosystem. *Carbon Balance and Management*, *12*(1), p.2.

- GDS has a complex land use (drainage) & disturbance (fire) history
- High natural methane fluxes from swamps and marshes
- Target for restoration
- Goal: Connect forest structure and species composition to methane and CO_2 fluxes

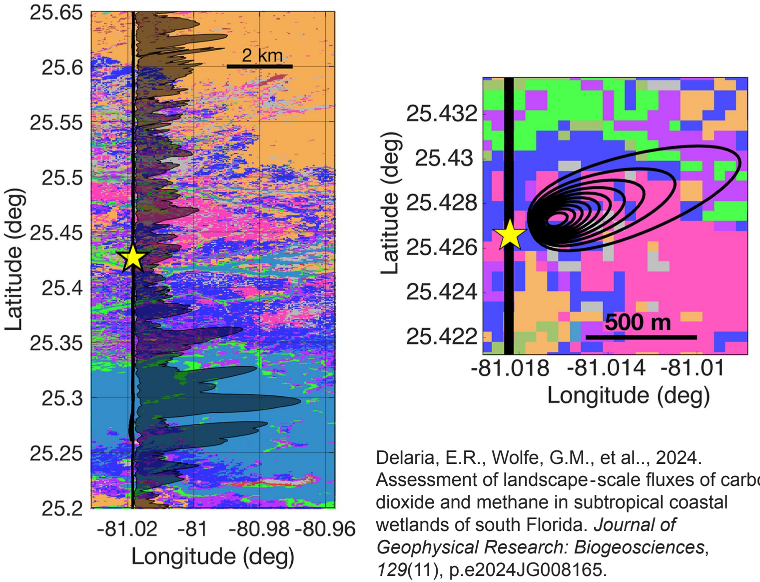
Science

Virginia Once Drained and Dried Peatlands, but Now Eyes Them as Carbon Sinks

The bogs, logged for centuries and recently burned, sequester far more carbon than forests. Now, thousands of acres are being "rewetted" as part of a restoration strategy.

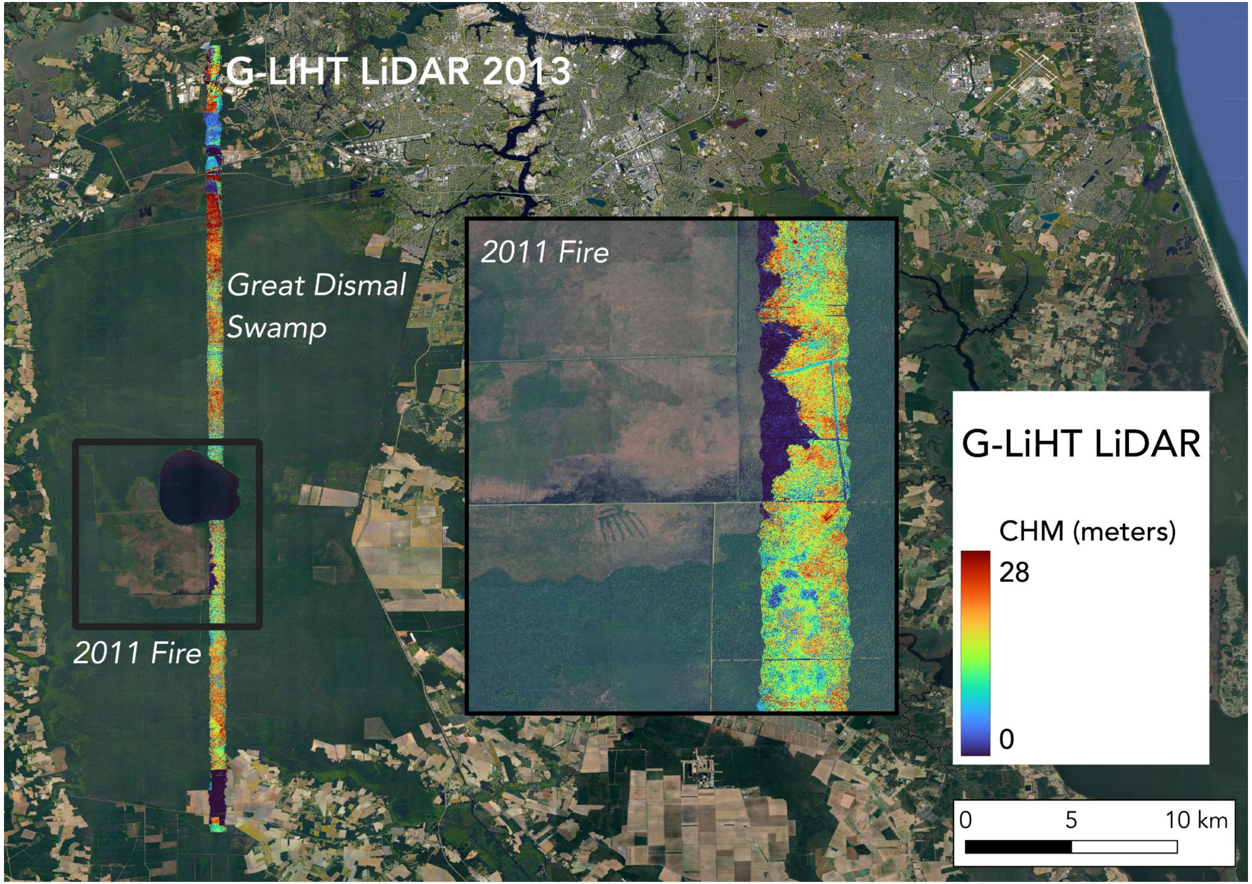
By Diana Kruzman
January 19, 2025

Inside Climate News

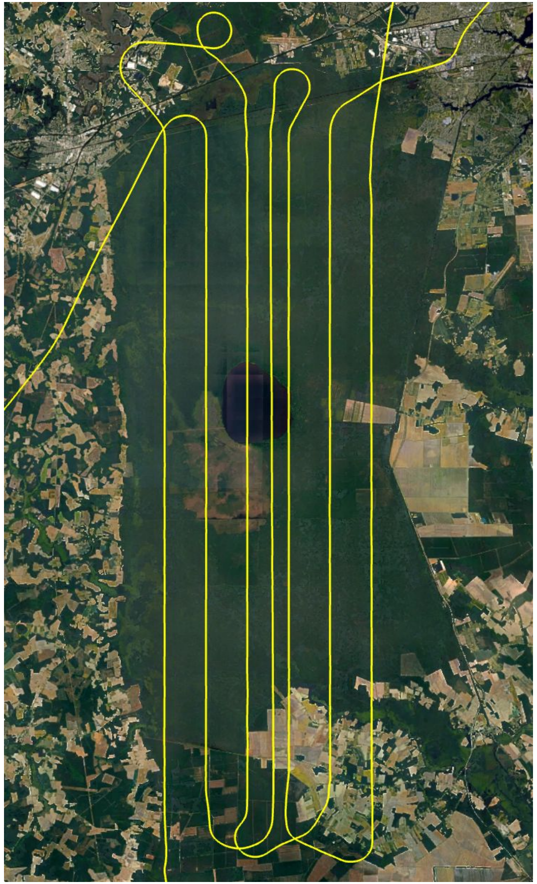


Delaria, E.R., Wolfe, G.M., et al.., 2024. Assessment of landscape-scale fluxes of carbon dioxide and methane in subtropical coastal wetlands of south Florida. *Journal of Geophysical Research: Biogeosciences*, *129*(11), p.e2024JG008165.

Great Dismal Swamp - Ecosystem Recovery

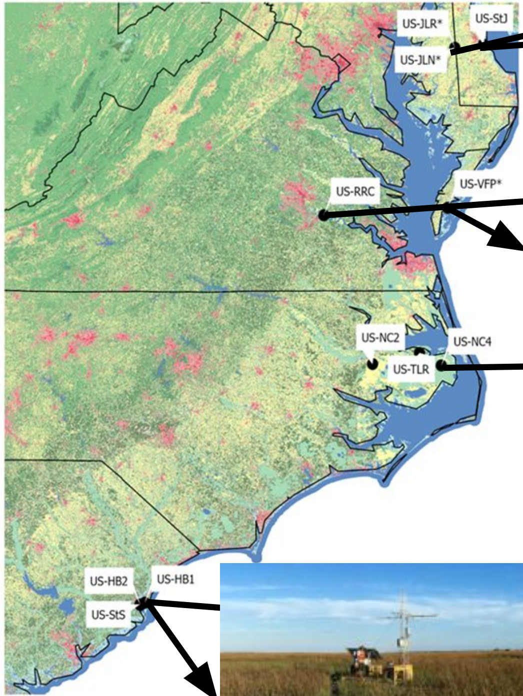


2025 G-LiHT Tracks

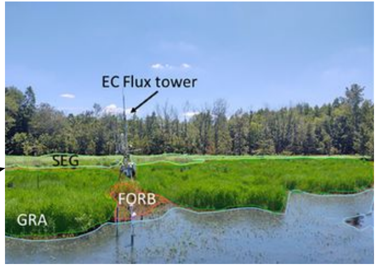


MidFlux Towers

AmeriFlux

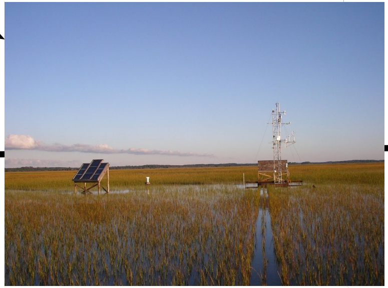


US-StJ



US-JLN

US-RRC



US-VFP

US-NC4

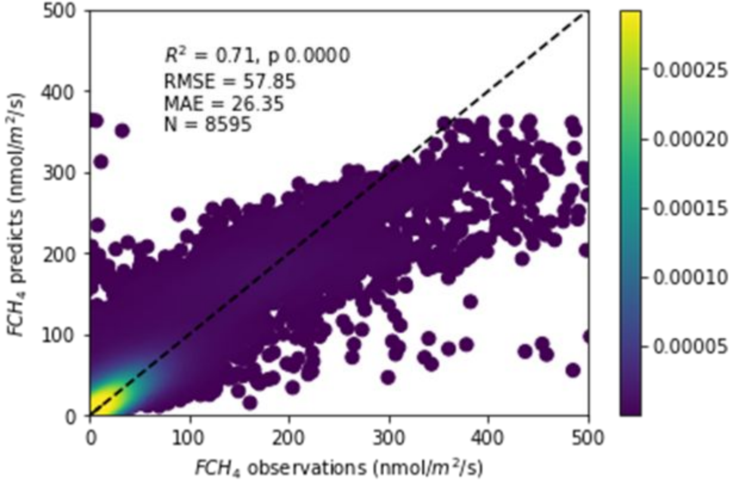
◀US-HB1

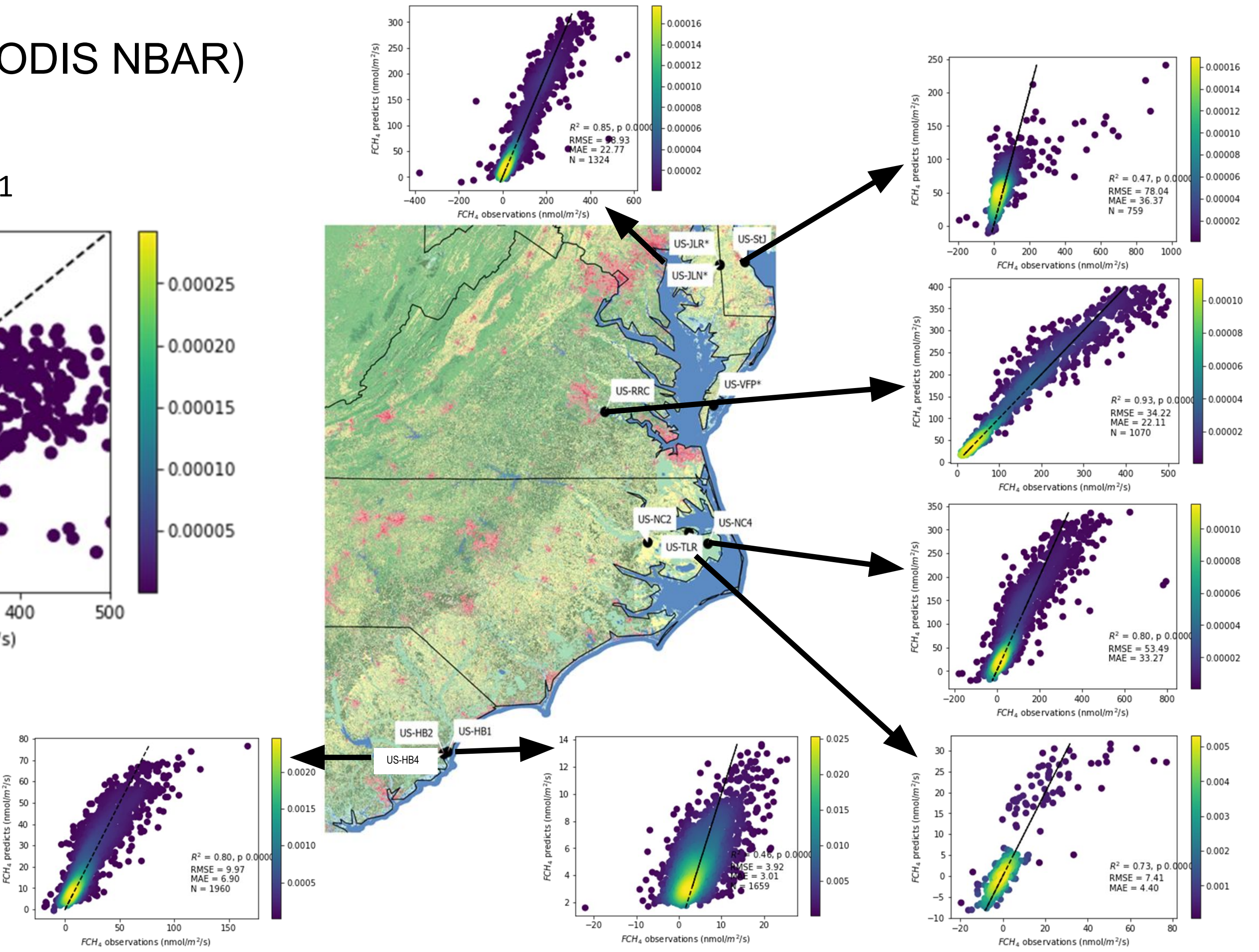
◀US-HB4

Slide courtesy Lisa Haber, VCU

MidFlux (FCH$_4$ vs MODIS NBAR)

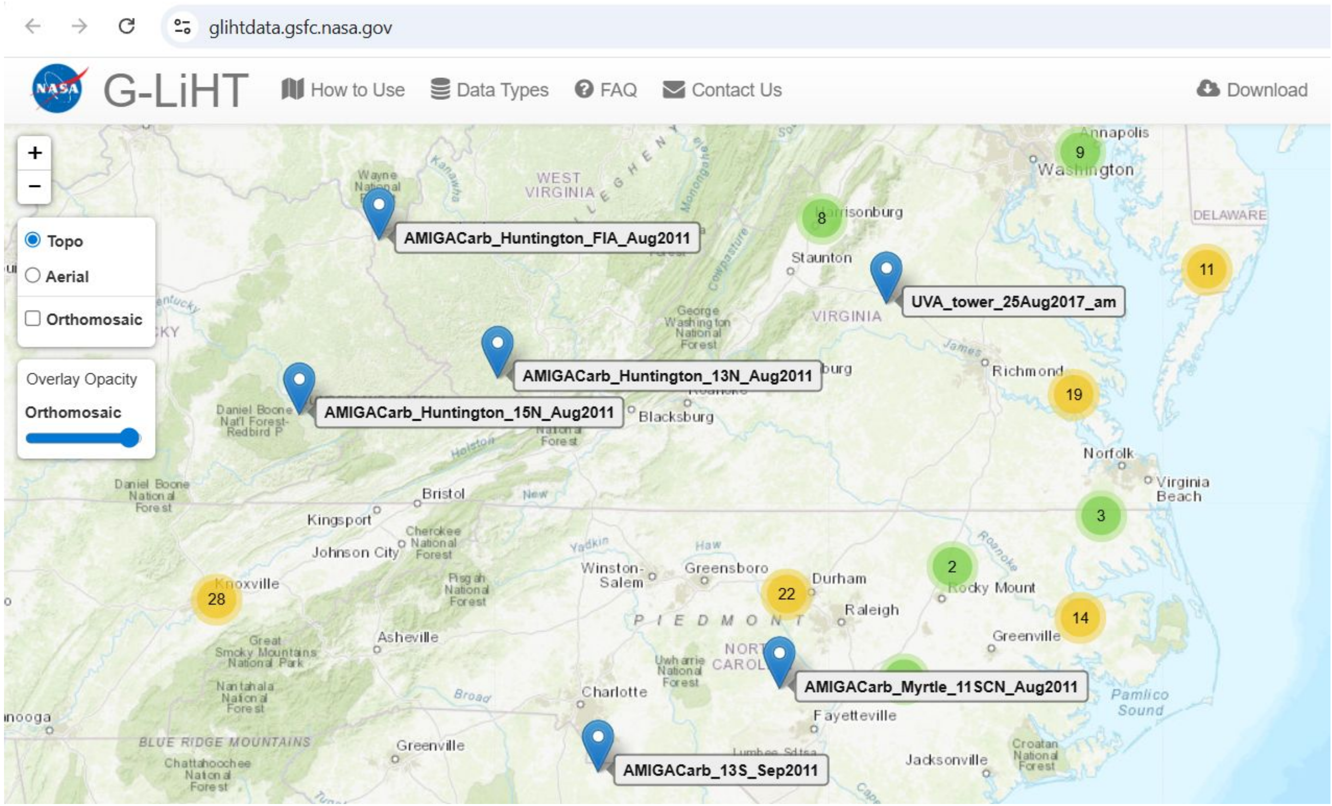
R^2 range: [0.46, 0.93];
pooled model R^2 = 0.71





Slide courtesy Lisa Haber (VCU) and Qing Ying (UMD)

Archive of Past Flights in Mid Atlantic Region



Other Past Campaigns

- Post-Hurricane damage assessment: *South Florida (2017), Puerto Rico (2018)*
- Coastal Ecosystems: G-LiHT airborne campaigns in *MA, FL Everglades and Bahamas (2014-17)*
- Forest Health: Joint NASA-USFS surveys for Forest Health in *NH, MA, NY and RI (2014-17)*
- Forest Inventory: NASA-USFS inventory of *Interior AK (2014, 2018)*
- Fluorescence: Joint NASA-ESA campaigns (*2013 FLEX-US; 2017 FLARE)*
- Landsat Calibration/Validation: G-LiHT and ground measurements in *OR, CA, and AZ (2013, 2015)*
- Carbon Cycle Science/AMIGA-Carb: G-LiHT airborne acquisitions in *CONUS and Mexico (2011-13)*
- Carbon Monitoring System: G-LiHT airborne acquisitions in *NC, MD, VA, WI, MA, ME (2011-17)*

Data Products

Initial standard data products (May 2026): trajectories, quicklooks, tile boundaries, classified LAZ pointclouds, canopy height model, digital terrain model, Lidar metrics, high-resolution Red, Green, Blue, and Near-Infrared (RBG-NIR) orthomosaics

Forthcoming products (by end of 2026): thermal infrared (TIR) mosaics and land surface temperature (LST), VSWIR surface reflectance and standardized vegetation index (SVI)

This image shows the combined G-LiHT instrument package and experimental Compact Fire Imager (CFI) onboard the A90

Partners and Collaborators